Alexander J. Elliott

alexjelliott.ca • +1 (647)-201-9466 • AlexJaElliott@gmail.com • linkedin.com/in/alexjaelliott

Video Editing • Robotics • Guitar • Photography • Physical AI • Climbing • Geography

EXPERIENCE

Magna International • Mechatronics CO-OP

Jan-Aug 2025 *Newmarket, ON*

Key Skills: *Engineering Software • Python • TypeScript • Simulation Automation • Data Visualization*

- Developed internal Python and TypeScript applications to automate pre- and post-processing of FEA, multibody dynamics, and inertial analysis, reducing repetitive engineering effort across global CAE teams
- Architected and deployed Inertia+, a software platform that automated static inertia calculations for a car door latch product and engineering report generation, reducing engineering effort by up to 26% (270+ hours saved per quarter)
- Led development of Collab Space, a web application built with JavaScript and Python for interactive FEA visualization

AME Group • QA/QC CO-OP

May-Aug 2024 *Vancouver, BC*

Key Skills: *Engineering Automation • Building Consulting • Python • C# • AutoCAD • Revit API • Process Improvement*

- Automated drafting and engineering workflows using Python and PyAutoCAD, as part of an effort to modernize company processes and reduce repetitive work within AutoCAD
- Developed C# Macros to automatically create schematics from existing BIM/MEP models and convert AutoCAD details

UBC Uncrewed Aircraft Systems • Mechanical Payload Team

Sep 2022–Sep 2024 *Vancouver, BC*

Key Skills: *UAV Systems • Mechanical Design • Payload Integration • Composite Manufacturing • Rapid Prototyping*

- Designed and prototyped payload mechanisms to safely drop water bottles, such as a winch system using electromagnetic clutches and rotary viscous damping, and guided parachutes
- Designed and manufactured aircraft structures and integrated payload systems using carbon fibre and 3D printing
- Helped achieve 2nd place (Phase 1) and 3rd place (Phase 2) at the 2024 AEAC competition

LRA3D • Mechanical CO-OP

May-Aug 2023 *Stouffville, ON*

Key Skills: *Industrial Engineering • CNC Machinery • Props and Sets • EtherCAT • Mach4*

- Scripted presets for CNC Routers and Hotwire machines using Mach4 to increase machine efficiency
- Wired and debugged CNC machines using EtherCAT to modernize production, enabling sub-millisecond control loops

FIRST Robotics Team 610 • Lead Programmer

October 2017-April 2021 *Toronto, ON*

Key Skills: *Controls Engineering • Robot Programming • Java • Prototyping • Computer Vision • Autonomous Control*

- Designed robots to compete in Power Up, Destination: Deep Space, and Infinite Recharge games, winning 2x District Event Winner, 4x Excellence in Engineering Award, 2x Industrial Design Award, 1x Championship Subdivision Finalist
- Led programming initiatives such as creating PoC work for swerve drive controls and integrating Limelight smart-camera for improved computer vision to enable better autonomous control

PROJECTS

Litter Collecting Drone • UBC Mech 458 Capstone

Sep 2025-Apr 2026

Key Skills: *Robotics • Computer Vision • YOLOv8 • Linux • ARM • Raspberry Pi OS • UAV • Embedded AI • V4L2*

- Designed and manufactured a quadcopter to autonomously detect and retrieve litter in difficult terrain
- Brought up and configured a Raspberry Pi Zero 2 W (Linux) for onboard trash inference, running YOLOv8 + OpenCV
- Integrated a Pixhawk flight controller with the companion computer over MAVLink/UART

Automated Spice Dispensing Robot • UBC Mech 423 Final Project

Jan-Apr 2026

Key Skills: *Embedded Systems • Robotics • Motor Control • PCB Design • Sensors • C • C#*

- Designed and developed a robotic spice dispensing system integrating embedded electronics and desktop software
- Wrote bare-metal C MSP430 firmware: interrupt-driven motor control, sensor polling (IR), and limit-switch handling
- Developed a C# interface supporting voice control, automated dispensing, and inventory management
- Built a UART serial protocol to communicate between the MCU and the C# app over USB

EDUCATION

University of British Columbia • BASc Mechanical Engineering

2021-2026 *Vancouver, BC*

BASc Mechanical Engineering • Mechatronics Option • w/ Co-Op • w/ Distinction • 5x Dean's List

SKILLS

Robotics • Embedded Systems • Computer Vision • Motor Control • Sensors & Instrumentation • UAV Systems

ML & Software • YOLOv8 • OpenCV • TensorFlow • React • SolidWorks • Onshape • I2C • UART • Linux • V4L2 • ARM

Programming • Python • C • C# • Java • MATLAB • Codex **Hardware** • Raspberry Pi • Arduino • ESP32 • PCB Design